

Suraj Shukla

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Full-stack software engineer with 5+ years building scalable web platforms, distributed systems, and applied machine learning in production. Fluent across the stack in TypeScript, Python, and Rust. Open to remote software roles.

PROFESSIONAL EXPERIENCE

Bramer Private Limited, Co-founder & CTO

2023 to 2026

New Delhi, India

- Set technical strategy, system architecture, and product roadmap; drove prioritization and scope from customer discovery through live pilot deployments.
- Built the full product stack as founding engineer: cross-platform ground control, an event-driven backend, and edge ML inference on embedded NVIDIA Jetson hardware, optimized for scalable deployment.
- Owned engineering process and product operations across customer pilots, on-site delivery, integration, and post-deployment iteration through first release and scaling.

PROJECTS

NirmaanOS

nirmaanos.vercel.app | github.com/beingglitch/nirmaanos

Multi-tenant SaaS with RBAC and event-sourced mutations. Next.js 16, React 19, Prisma 7, PostgreSQL

- Designed a dependency-graph computation engine where node values resolve through several bases with both top-down and bottom-up edges; a single topological pass diverges, so I implemented fixed-point iteration that converges any acyclic graph in under 30 passes.
- Hardened the write path for connection-pooled PostgreSQL: idempotent recompute plus dirty-row diffing in concurrent batches removes long transactions, with copy-on-write versioning and per-node lineage powering cross-version diffs.

fabricOS

app.bramer.in | github.com/beingglitch/fabricos

Multi-robot ground control and sensor analytics. Rust, TypeScript, Tauri 2, MAVLink

- Architected a single Rust hub-core compiled to three runtimes via Cargo feature flags: headless Axum server, Tauri 2 desktop, and Tauri 2 Android, with zero-port desktop embedding (a custom scheme handler pipes HTTP into the Axum router; Tauri events replace WebSockets).
- Built an event-driven core on Redis Streams with real-time telemetry fan-out to 50+ clients, and designed a plugin microkernel: crash-isolated plugins over a typed sensor, process, computer-vision, and insight pipeline for LIDAR, Ground-Penetrating Radar (GPR), and thermal data, with WebAssembly sandboxing.

LEADERSHIP & RESEARCH ROLES

I-Hub Foundation for Cobotics (IHFC), IIT Delhi, Research Fellow

2026

Delhi, India

- Selected for a competitive fellowship in collaborative robotics at IHFC, a Department of Science and Technology Innovation Hub at IIT Delhi; building a swarm-coordination system for multi-agent robotics, drone-first and designed to generalize to any robot platform.

DTU IoT Lab, Head of Development

Jun 2025 to Present

Delhi, India

- Lead a team of student developers across the lab's drone and sensor-network research; own delivery on software-defined networking (SDN) and resilient mesh-relay workstreams.

DTU Unmanned Aerial Systems (UAS) Team, Software & Autonomy Lead

Aug 2022 to Dec 2023

Delhi, India

- Led the software and autonomy sub-team; delivered the GPS-denied navigation stack (Robot Operating System 2, ROS 2), an early swarm-coordination framework, and the onboard visual perception pipeline for navigation.

SKILLS

Languages: Python, TypeScript, JavaScript, Go, Java, Kotlin, Rust, C, C++, SQL, Bash

Frontend: React, Next.js, React Native, Flutter, TanStack, Zustand, Tauri 2, TailwindCSS

Backend: Node.js, FastAPI, Axum, Tokio, REST, WebSocket, WebRTC

Data: PostgreSQL, PostGIS, MongoDB, Redis, Firebase, Prisma, SQLite

Cloud and Infrastructure: AWS, GCP, Docker, GitHub Actions (self-hosted runners), Vercel, Cloudflare, pnpm, Turborepo

AI and Machine Learning: PyTorch, OpenCV, LangGraph, LangChain

RESEARCH

- **Neuromorphic SLAM:** chip-based module (under 5 W) using event-camera perception for real-time spatial mapping on micro-swarm drones in GPS-denied environments.
- **Subsurface Ground-Penetrating Radar (GPR):** radar signal processing and machine-learning B-scan segmentation to reconstruct subsurface structures.

EDUCATION

- **Delhi Technological University (DTU)** 2022 to 2026
B.Tech in Software Engineering Delhi, India